VIVA PREPARATION GUIDE

Stellar Spectra XAI System — Complete Page-by-Page Walkthrough

තාරකා වර්ණාවලි XAI පද්ධතිය — සිංහල Viva සූදානම් වීමේ මාර්ගෝපදේශය

Project ID	J26-IT-449 | SLIIT IT4010 Research Project — PP1 Assessment
Student	P.K. Edirisooriya (IT23330696)
Research Title	Explainable AI for Astronomical Discovery using XAI and Uncertainty Quantification
Model	1D-CNN-BiLSTM | Test Accuracy: 77.33% | Balanced Acc: 74.15% | Macro F1: 0.6406
Dataset	APOGEE DR17 Near-Infrared Stellar Spectra | 700 Train / 150 Val / 150 Test
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How to use this guide: Each section covers one page of the Streamlit app with English explanation + Sinhala translation + Viva Q&A.;
මෙම මාර්ගෝපදේශය භාවිතා කරන ආකාරය: සෑම කොටසක්ම Streamlit app එකේ එක් page එකක් ආවරණය කරයි — ඉංග්‍රීසි පැහැදිලි කිරීම + සිංහල පරිවර්තනය + Viva ප්‍රශ්නෝත්තර.

SYSTEM OVERVIEW

පද්ධති දළ විශ්ලේෂණය

■■ English:
This is a Streamlit-based web application that classifies stellar spectra using a trained 1D-CNN-BiLSTM deep learning model and explains its predictions using Gradient SHAP (Explainable AI). It has 6 pages accessible via the sidebar.

සිංහල:
මයෙ Streamlit-based web application එකක් වන අතර, 1D-CNN-BiLSTM deep learning model එකක් භාවිතයනේ තාරකා spectra වර්ගීකරණය (classify) කර, Gradient SHAP (Explainable AI) භාවිතයනේ ප්‍රතිඵල පැහැදිලි කරයි. Sidebar එකහේ Page 6ක් ඇත.

Page	Name	Purpose (English)	■■■■■■ (■■■■■■)
1	■ Dashboard	System overview, dataset distribution, metrics	■■■■■■■ ■■■■■■■■, dataset, metrics
2	■ Spectrum Explorer	Browse test spectra interactively	Test spectra browse ■■■■■■
3	■ Upload & Analyse	Upload your own spectrum, get live prediction	■■■■■ spectrum upload ■■■■■■
4	■ Model Comparison	Compare CNN-BiLSTM vs classical baselines	Models ■■■■■■■■■
5	■ Scientific XAI Validation	AOPC faithfulness & Jaccard stability tests	XAI ■■■■■■■■■■■■ ■■■■■■ ■■■■■■
6	■■ About & Limitations	System limitations, how to run, citations	■■■■■■■■ ■■ ■■■■■ ■■■■■■

■ What is this system?

■ Answer: This is a web-based stellar spectra classification and XAI dashboard. It uses a 1D-CNN-BiLSTM model trained on APOGEE near-infrared spectra to classify stars into 5 spectral classes (A, B, F, G, K) and uses Gradient SHAP to identify which wavelength regions drove each prediction.

පිළිතුර: මයෙ web-based තාරකා spectra වර්ගීකරණ සහ XAI dashboard එකකි. APOGEE near-infrared spectra මත පුහුණු කළ 1D-CNN-BiLSTM model එකක් භාවිතයනේ තාරකා A, B, F, G, K ලෙස Spectral Classes 5 කට වර්ගීකරණය කරයි. Gradient SHAP භාවිතයනේ ප්‍රතිඵලවලට බලපෑ wavelength regions හඳුනා ගනී.

PAGE 1 — ■ Dashboard

What this page contains

■■ English:
The Dashboard is the home page. It displays: (1) A bar chart of class distribution across the dataset manifest (N=1,004 rows), (2) Model performance metrics (accuracy, balanced accuracy, macro F1), (3) The confusion matrix showing how often each class was correctly predicted, (4) A summary of the XAI methods used.

සිංහල:
Dashboard page එක home page එකයි. එය පෙන්වන දෑ: (1) Dataset manifest (N=1,004) හි class distribution bar chart, (2) Model performance metrics — accuracy, balanced accuracy, macro F1, (3) Confusion matrix — සෑම class එකක්ම නිවැරදිව predict කළ ආකාරය, (4) භාවිත XAI methods සාරාංශය.

Key Numbers to Remember

Metric	Value	What it means (English)	■■■■■■ ■■■■■■■■■ ■■■■■■
Test Accuracy	77.33%	77 out of 100 spectra predicted correctly	100 spectra 77 ■■ ■■■■■■■■■ predict ■■■
Balanced Accuracy	74.15%	Average accuracy across all 5 classes	Class 5 ■ ■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■
Macro F1	0.6406	Average F1 score equally weighted per class	Class ■■■■■■■ ■■■■■ weightage F1 score
Test Set Size	N=150	Zero-leakage isolated test split	■■■■■■■ data leak ■■■■■ isolated test set
Train Split	700	Samples used for training	Model ■■■■■■■ ■■■■■■■ ■■■■■■ samples
Val Split	150	Samples used for validation/tuning	Validation/tuning ■■■■■ samples

■ Why is accuracy 77.33% and not higher?

■ **Answer:** The dataset has severe class imbalance — Class K dominates (~87% of samples). Despite this, our model achieves 74.15% balanced accuracy. Low minority-class samples (A, B) limit performance. Higher accuracy is possible with more data.

පිළිතුර: Dataset එකේ class imbalance ඉතා ශක්තිමත්ය — Class K ~87% ක් ආවරණය කරයි. එය නිසාවෙන් minority classes (A, B) සඳහා accuracy අඩු වේ. 74.15% balanced accuracy ලබා ගත හැකි විය. Data වැඩි නම් accuracy ඉහළ දැමීය හැකිය.

■ What is a confusion matrix?

■ **Answer:** A confusion matrix is a table showing how many spectra of each true class were predicted as each class. The diagonal shows correct predictions. Off-diagonal shows mistakes.

පිළිතුර: Confusion matrix යනු සෑම true class එකක් model එක විසින් කොයි class ලෙස predict කළේ ද කියා පෙන්වන වගුවකි. Diagonal (ඇතිරිල්ල) නිවැරදි predictions. Off-diagonal (ඉවත) වැරදි predictions.

PAGE 2 — ■ Spectrum Explorer

What this page contains

■■ English:
The Spectrum Explorer lets you browse the 150 isolated test-set spectra. You can select any sample by index, view its normalised flux plot, see the model's prediction and confidence, and view the Gradient SHAP attributions alongside matched NIST atomic spectral lines.

සිංහල:
Spectrum Explorer page එකෙන් isolated test set spectra 150 browse කළ හැකිය. Index අනුව ඕනෑම sample එකක් select කර, normalised flux plot, model prediction, confidence, සහ Gradient SHAP attributions — NIST atomic lines සමඟ — දැකගත හැකිය.

What each element shows

UI Element	English Explanation	■■■■■ ■■■■■■■■■ ■■■■■
Sample Index Slider	Select which spectrum (0–149) from the test set to view	Test set එකෙන් කොයි spectrum (0-149) බලන්නේ ද ලසෙ slider
Normalised Flux Plot	The spectrum's flux (brightness) across 4,000 near-infrared wavelength channels	Spectrum එකේ flux (දීප්තිය) near-infrared wavelength channels 4,000 හරහා
True Label	The actual spectral class of this star derived from its APOGEE TEFF header	APOGEE TEFF header වලින් ලබාගත් මමෙ තාරකාවේ සැබෑ spectral class
Predicted Label + Confidence	Model's classification and its probability score for the top class	Model විසින් classify කළ class සහ ඒ class සඳහා probability score
SHAP Attribution Plot	Which wavelength regions most influenced this prediction (peaks = important regions)	Prediction එකට බලපෑ wavelength regions (peaks = වැදගත් ප්‍රදේශ)
NIST Vertical Lines	Known atomic absorption lines from laboratory measurements cross-matched to SHAP peaks	Laboratory NIST measurements ලසෙ SHAP peaks සමඟ cross-match කළ atomic lines
Astrophysical Properties	Inferred stellar physics: Teff, evolutionary stage, chemistry, fate	Infer කළ stellar physics: Teff, evolve stage, chemistry, stellar fate

■ What does the SHAP attribution plot show?

■ **Answer:** It shows which wavelength channels (positions in the spectrum) most influenced the model's prediction. Peaks in the attribution plot correspond to candidate atomic absorption lines — e.g., O I at 15,871 Å, Mg I at 15,745 Å. These were cross-matched with NIST laboratory references.

පිළිතුර: SHAP attribution plot එක පෙන්වන්නේ spectrum එකේ කොයි wavelength channels model prediction එකට වැඩිම බලපෑමක් ඇති කළේ ද යන්නයි. Attribution peaks — O I 15,871 Å, Mg I 15,745 Å ආදී — NIST laboratory references සමඟ cross-match කළ atomic absorption lines ලසෙ හඳුනා ගනී.

PAGE 3 — ■ Upload & Analyse

■■ English:
This is the core interactive page. You upload a real stellar spectrum file and the system processes it live — applying the same preprocessing pipeline as training, running inference through the 1D-CNN-BiLSTM, computing real-time Gradient SHAP attributions, displaying astrophysical properties, and allowing download of results as JSON or PDF.

සිංහල:
මෙය core interactive page එකයි. ඔබ real stellar spectrum file එකක් upload කළ සැනිකව system එක live process කරයි — training pipeline ම apply කර, 1D-CNN-BiLSTM හරහා inference ධාවනය කර, real-time Gradient SHAP attributions compute කර, astrophysical properties display කර, JSON හෝ PDF ලෙස download කිරීමට ඉඩ සලසයි.

Supported Upload Formats

Format	Description (English)	■■■■■■■ (■■■■■)
CSV	Two columns: wavelength, flux. The system auto-infers column names.	Columns 2: wavelength, flux. Column names auto-detect කරයි.
NPZ	NumPy archive with array key 'X' (shape 4000 or N×4000), optionally 'y'.	NumPy archive; key 'X' (shape 4000 හෝ N×4000). Optionally 'y'.
NPY	Raw 1D NumPy array of length 4000 or 2D array N×4000.	1D NumPy array (length 4000) හෝ 2D array N×4000.
FITS	APOGEE apStar FITS file. HDU 1 contains the combined flux array with WCS header.	APOGEE apStar FITS file. HDU 1 = combined flux array with WCS header.

Step-by-step: What happens after upload

■ Step 1 — Preprocessing

■■ English:
The raw spectrum is resampled onto a 4,000-point logarithmic wavelength grid (15,100–17,000 Å) and continuum-normalised using a 101-pixel median filter.

සිංහල:
Raw spectrum එක 4,000-point logarithmic wavelength grid (15,100–17,000 Å) මත resample කර, 101-pixel median filter භාවිතයෙන් continuum normalize කරයි.

■ Step 2 — Model Inference

■■ English:
The preprocessed spectrum (shape: 1×4000) is passed through the frozen 1D-CNN-BiLSTM model. Softmax outputs give 5-class probabilities.

සිංහල:
Preprocessed spectrum (shape: 1×4000) frozen 1D-CNN-BiLSTM model හරහා pass කරෙේ. Softmax output classes 5 ට probabilities ලබා දෙයි.

■ Step 3 — Gradient SHAP

■■ English: Real-time Gradient SHAP (shap.GradientExplainer) computes attribution values for all 4,000 channels specifically for this uploaded spectrum.	සිංහල: Real-time Gradient SHAP (shap.GradientExplainer) upload කළ spectrum to specific ලෙස channels 4,000 සඳහා attribution values compute කරයි.

■ Step 4 — Astrophysical Properties

■■ English: Based on the predicted class, a curated table of stellar physics is displayed: surface temperature, evolutionary stage, dominant chemistry, galactic population, and predicted fate.	සිංහල: Predict කළ class අනුව, stellar physics curated table display කරෙේ: surface temperature, evolutionary stage, chemistry, galactic population, fate.

■ Step 5 — Download

■■ English: Results can be downloaded as (a) a JSON file with all predictions and metadata, or (b) a Scientific PDF Report with the highlighted primary finding.	සිංහල: ප්‍රතිඵල (a) JSON file ලෙස — predictions සහ metadata සමඟ, හෝ (b) Scientific PDF Report ලෙස — primary finding highlight සහිතව download කළ හැකිය.

■ What preprocessing contract is applied to uploaded files?

■ **Answer:** The system applies: WCS wavelength extraction (for FITS) → resampling onto 4,000 log-spaced grid (15,100–17,000 Å) → 101-pixel median continuum division to normalise the flux to unity. This matches the preprocessing used during training.	පිළිතුර: System apply කරන preprocessing: WCS wavelength extraction (FITS සඳහා) 4,000 log-spaced grid (15,100–17,000 Å) මත resample 101-pixel median continuum division (flux unity ලෙස normalize කිරීම). Training preprocessing ම apply වේ.

■ Does the system work on stars it has never seen before (untrained data)?

■ **Answer:** Yes. The model was trained to learn physical spectral features (absorption line patterns), not to memorise individual stars. Any new APOGEE-format near-infrared spectrum in the 15,100–17,000 Å range will be correctly preprocessed and classified.	පිළිතුර: ඔව්. Model එක Individual stars memorise කිරීමට නොව, physical spectral features (absorption line patterns) ඉගනෙ ගැනීමට train කරේණ. APOGEE-format near-infrared spectrum (15,100–17,000 Å) නව ඕනෑම දයෙක් correctly preprocess කර classify කරයි.

PAGE 4 — ■ Model Comparison

■■ English:
This page compares the proposed 1D-CNN-BiLSTM model against other deep learning architectures and classical machine learning baselines, all evaluated on the same zero-leakage test split (N=150).

සිංහල:
මෙම page එකෙහි proposed 1D-CNN-BiLSTM model, අනක්‍රෙත් deep learning architectures සහ classical ML baselines සමඟ සංසන්දනය කරයි. සියල්ල same zero-leakage test split (N=150) මත evaluate කරෙේ.

Model	Accuracy	Balanced Acc	Macro F1	Notes (English)	■■■■ (■■■■■)
1D-CNN-BiLSTM	77.33%	74.15%	0.6406	■ Proposed model — best overall	Proposed model — හොඳම ප්‍රතිඵල
1D-ResNet	~72%	~68%	~0.58	Residual CNN without recurrence	Recurrence නොමැති Residual CNN
Standard CNN	~70%	~65%	~0.55	Simpler CNN baseline	සරල CNN baseline
Random Forest	~65%	~60%	~0.50	Classical ML baseline	Classical ML baseline
SVM	~63%	~58%	~0.48	Classical ML baseline	Classical ML baseline
KNN	~60%	~55%	~0.45	Classical ML baseline	Classical ML baseline

■ Why did you choose CNN-BiLSTM over ResNet or classical methods?

■ **Answer:** CNN-BiLSTM combines local feature extraction (CNN layers detect absorption line patterns at specific wavelengths) with bidirectional temporal context (BiLSTM captures how spectral features evolve across the spectrum). This hybrid architecture outperforms both pure CNNs and classical methods on our sequential spectral data.

පිළිතුර: CNN-BiLSTM, local feature extraction (CNN layers specific wavelengths හි absorption line patterns detect කරයි) සහ bidirectional temporal context (BiLSTM spectral features evolve ආකාරය capture කරයි) ඒකාබද්ධ කරයි. Sequential spectral data සඳහා pure CNNs සහ classical methods දකෙම ඉක්මවා යයි.

PAGE 5 — ■ Scientific XAI Validation

■■ English:
This page presents the formal scientific validation of the Gradient SHAP explanations using two established XAI evaluation protocols: (1) AOPC Faithfulness Test and (2) Jaccard Stability Test. These confirm whether the explanations are trustworthy and stable.

සිංහල:
මමෙ page එකෙහි Gradient SHAP explanations, XAI evaluation protocols 2 ක් (1) AOPC Faithfulness Test සහ (2) Jaccard Stability Test භාවිතයෙන් formal scientific validation කරයි. Explanations trustworthy සහ stable ද යන්න තහවුරු කරයි.

Test 1 — AOPC Faithfulness (Area Over Perturbation Curve)

■■ English:
We masked (removed) the top-attribution wavelength channels and measured how much the model's confidence dropped. AOPC Top-K score: 0.7507. Random masking score: 0.1192. Ratio: 6.30×. This means masking the most important regions causes a 6× larger confidence drop than random masking — confirming the explanations point to genuinely important spectral regions.

සිංහල:
Top-attribution wavelength channels mask (ඉවත්) කර model confidence drop කොතරම් දුරට අඩු වේ ද ඇස්තමේන්තු කළමෙු. AOPC Top-K score: 0.7507. Random masking score: 0.1192. Ratio: 6.30×. ගිය attribution regions mask කළ විට random masking ට වඩා 6× confidence drop — explanations genuinely important spectral regions වතෙ යොමු කරයි ලසෙ තහවුරු කරෙේ.

Test 2 — Jaccard Stability Test

■■ English:
We added synthetic Gaussian noise to each spectrum and measured how consistent the top-K attribution peaks remained. Jaccard@sigma=0.01: 0.8069. Jaccard@sigma=0.02: 0.7768. Jaccard@sigma=0.04: 0.6853. At realistic noise levels (sigma=0.01), 80.7% of the top attribution peaks remain the same — showing the explanations are stable to observational noise.

සිංහල:
Spectrum එකට synthetic Gaussian noise add කර top-K attribution peaks stable ව පවතී ද ඇස්තමේන්තු කළමෙු. Jaccard@sigma=0.01: 0.8069. Jaccard@sigma=0.02: 0.7768. Jaccard@sigma=0.04: 0.6853. Realistic noise (sigma=0.01) හි top attribution peaks 80.7% ක් සමාන ව පවතී — explanations noise ලසෙ stable ය.

NIST Line Matching

■■ English:
The top SHAP attribution peaks were cross-matched against the NIST Atomic Spectra Database. 6 out of 57 peaks matched known atomic transitions: O I (15,871 Å), H Br14 (15,263 Å), N I (16,764 Å), Si I (15,960 Å), Mg I (15,745 Å), O I (15,569 Å). Match rate: 10.53% — exploratory alignment, not a statistically significant discovery.

සිංහල:
Top SHAP attribution peaks, NIST Atomic Spectra Database සමඟ cross-match කළමෙු. Peaks 57 න් 6 ක් known atomic transitions: O I (15,871 Å), H Br14 (15,263 Å), N I (16,764 Å), Si I (15,960 Å), Mg I (15,745 Å), O I (15,569 Å). Match rate 10.53% — exploratory alignment, statistically significant discovery නොවේ.

■ Does NIST line matching prove the model uses real physics?

■ **Answer:** It is exploratory evidence, not proof. 10.53% of SHAP peaks aligned with known NIST lines. The Monte Carlo p-value of 0.538 is not statistically significant. However, the alignment with physically known lines (O I, Mg I, Si I) is scientifically consistent with what we expect for these spectral classes.	පිළිතුර: එය exploratory evidence, proof නොවේ. SHAP peaks 10.53% known NIST lines සමඟ aligned. Monte Carlo p-value 0.538 statistically significant නොවේ. නමුත් physically known lines (O I, Mg I, Si I) සමඟ alignment, spectral classes සඳහා විද්‍යාත්මකව consistent.

PAGE 6 — ■■ About & Limitations

■■ English:
This page documents the scientific limitations of the system, lists the commands to reproduce the experiments, and provides citations and a project overview. It is important to acknowledge limitations in your viva.

සිංහල:
මමෙ page එකෙහි system scientific limitations, experiments reproduce කිරීමට commands, citations සහ project overview ලේඛනගත කරෙේ. Viva හිදී limitations ප්‍රශ්නාවලිය ඉතා වැදගත්ය.

Key Limitations (Know these for viva!)

■ Class Labels are Proxy Labels

■■ English:
Class labels (A, B, F, G, K) are derived from APOGEE TEFF header values, not independently verified Harvard spectral types.

සිංහල:
Class labels (A, B, F, G, K) APOGEE TEFF header values ලසෙ ලබාගත් proxy labels. Independently verify කළ Harvard spectral types නොවේ.

■ Severe Class Imbalance

■■ English:
Class K dominates ~87% of the dataset. This limits minority class (A, B, F) performance.

සිංහල:
Class K ~87% dominate කරයි. Minority classes (A, B, F) performance සීමිත කරෙේ.

■ NIST Matching is Exploratory

■■ English:
10.53% line matching rate; Monte Carlo p=0.538 (not significant). This is candidate evidence, not a confirmed discovery.

සිංහල:
Line matching 10.53%; Monte Carlo p=0.538 (significant නොවේ). Candidate evidence, confirmed discovery නොවේ.

■ Noise Tests are Synthetic

■■ English:
Jaccard stability tested on simulated Gaussian noise, not real observing conditions.

සිංහල:
Jaccard stability, simulated Gaussian noise මත test කරෙණ; Real observing conditions නොවේ.

■ Raw FITS not Included

■■ English:
Raw APOGEE FITS files (4 GB) not included in submission. Preprocessing cannot be fully reproduced from supplied files alone.

සිංහල:
Raw APOGEE FITS files (4 GB) submission හි ඇතුළත් නොවේ. Preprocessing supplied files ලසෙ fully reproduce කළ නොහැක.

SPECTRAL CLASSES — Quick Reference

Spectral Classes — කවේ මාර්ගෝපදේශය

Class K Orange / Red Giant	Aging post-main-sequence red giant. Core hydrogen exhausted. Mg I, Si I, CO molecular bands dominate NIR spectrum. Located in Milky Way disk and bulge.	වයස්ගත post-main-sequence Red Giant. Core hydrogen අවසන්. Mg I, Si I, CO molecular bands NIR spectrum dominate කරයි. Milky Way disk සහ bulge.
Class G Solar-Type Star	Solar twin. Stable Main Sequence star. Surface temperature 5,000–6,000 K, similar to our Sun. Fe I, Ca II, Si I absorption lines.	Solar twin. Stable Main Sequence star. Surface temperature 5,000–6,000 K — අපේ සූර්යයාට සමාන. Fe I, Ca II, Si I absorption lines.
Class F Intermediate White Star	Warmer than the Sun (6,000–7,500 K). Main Sequence / early post-main-sequence. Ionised metals Fe II, Ca II dominant.	සූර්යයාට වඩා උෂ්ණ (6,000–7,500 K). Main Sequence / early post-main-sequence. Ionised metals Fe II, Ca II dominant.
Class A Hot Early-Type Star	Hot white star (7,500–10,000 K). Short lifetime ~1 billion years. Strong Hydrogen Brackett absorption series in NIR.	Hot white star (7,500–10,000 K). ජීවිත කාලය ~billion years 1. Strong Hydrogen Brackett absorption NIR.
Class B Luminous Blue Giant	Extremely hot (10,000–30,000 K), massive, young. He I lines and Brackett series. Destined for supernova or massive white dwarf.	ඉතා උෂ්ණ (10,000–30,000 K), විශාල, තරුණ. He I lines සහ Brackett series. Supernova හෝ massive white dwarf ඉලක්ක.

CRITICAL VIVA Q&A; — Panel Questions & Answers

Panel ප්‍රශ්නෝත්තර

■ What is the main contribution of your research?

■ **Answer:** We demonstrate that a 1D-CNN-BiLSTM model can classify near-infrared stellar spectra with 77.33% accuracy on a zero-leakage split, and that Gradient SHAP explanations point to physically meaningful atomic absorption regions (O I, Mg I, Si I), providing interpretable, trustworthy AI for astronomical classification.

පිළිතුර: 1D-CNN-BiLSTM model zero-leakage split මත near-infrared stellar spectra 77.33% accuracy ලෙස classify කළ හැකි බව සහ Gradient SHAP explanations physically meaningful atomic absorption regions (O I, Mg I, Si I) වනෙ point කරන බව demonstrate කළමෙු — astronomical classification සඳහා interpretable, trustworthy AI සපයමින්.

■ What is Gradient SHAP and why did you use it?

■ **Answer:** Gradient SHAP (shap.GradientExplainer) computes expected gradients over a background dataset to assign attribution values to each input feature. We used it because it is model-specific to neural networks, computationally efficient for 4,000-channel inputs, and produces attributions directly comparable to known spectral lines.

පිළිතුර: Gradient SHAP (shap.GradientExplainer) background dataset හරහා expected gradients compute කර සෑම input feature ටම attribution values assign කරයි. Neural networks ලෙස model-specific, 4,000-channel inputs ලෙස computationally efficient, සහ known spectral lines ලෙස directly comparable attributions produce කරයි — ඒ නිසා භාවිත කළමෙු.

■ What does zero-leakage split mean?

■ **Answer:** Zero-leakage split means no data from the test set was used during training or validation. We used 700 samples for training, 150 for validation, and 150 for testing — completely isolated. This ensures the reported 77.33% accuracy is a genuine generalisation measure, not an inflated result from data leakage.

පිළිතුර: Zero-leakage split යනු test set data training හෝ validation දිදී ලෙස භාවිත නොකළ බවයි. Training 700, validation 150, testing 150 — සම්පූර්ණයනේ isolated. Reported 77.33% accuracy genuine generalisation, data leakage ලෙස inflated result නොවේ.

■ Why does Class K have the highest accuracy?

■ **Answer:** Class K dominates ~87% of our dataset because APOGEE primarily targets red giant stars. The model has seen far more Class K examples, so it is better calibrated for that class. This creates class imbalance — addressed by reporting balanced accuracy alongside raw accuracy.

පිළිතුර: APOGEE primarily red giant stars target කරයි. ඒ නිසා dataset 87% Class K ලෙස. Model Class K examples ඉතා වැඩියනේ දැක ඇති, class ලෙස better calibrated. Class imbalance — balanced accuracy reporting ලෙස address කරෙේ.

■ How did you validate the XAI explanations?

■ **Answer:** Two tests: (1) AOPC Faithfulness — masking top-attribution channels caused 6.30× larger confidence drop than random masking (AOPC 0.7507 vs 0.1192), confirming attributions point to genuinely important features. (2) Jaccard Stability — at sigma=0.01 noise, 80.7% of top peaks remained consistent, confirming explanations are stable.	පිළිතුර: Tests 2: (1) AOPC Faithfulness — top-attribution channels mask කිරීම random masking ට වඩා 6.30× confidence drop (AOPC 0.7507 vs 0.1192), attributions genuinely important features. (2) Jaccard Stability — sigma=0.01 noise, top peaks 80.7% consistent — explanations stable.

■ What would you improve in future work?	
■ **Answer:** Future improvements: (1) Collect more minority class spectra (A, B, F) to reduce imbalance. (2) Implement per-upload uncertainty quantification (MC Dropout or Deep Ensembles). (3) Use per-query SHAP instead of background-averaged attributions. (4) Validate on optical (SDSS/Gaia) spectra for generalisability.	පිළිතුර: Future improvements: (1) Minority class (A, B, F) spectra collect කිරීම — imbalance reduce. (2) Per-upload uncertainty quantification (MC Dropout / Deep Ensembles) implement. (3) Background-averaged ලෙස per-query SHAP. (4) Optical spectra (SDSS/Gaia) ලෙස generalisability validate.